

1. Create a package com.math.operations and write a class to perform addition and subtraction.

Code:

```
package com.math.operations;

public class ArithmeticOperations {

    public static int add(int a, int b) {

        return a+b;

    }

    public static int sub(int a, int b) {

        if(a > b) {

            return a-b;

        }

        else {

            return b-a;

        }

    }

    public static void main(String[] args) {

        int x = 15;

        int y = 20;

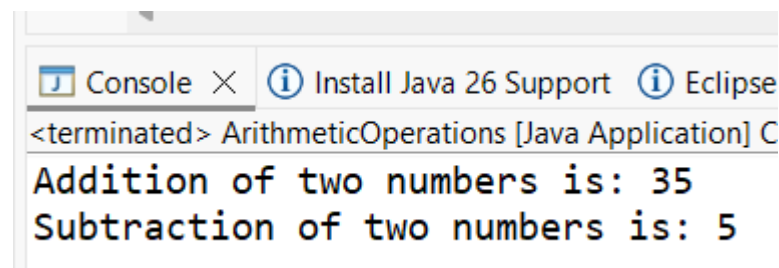
        System.out.println("Addition of two numbers is: " + add(x,y));

        System.out.println("Subtraction of two numbers is: " + sub(x,y));

    }

}
```

Output:

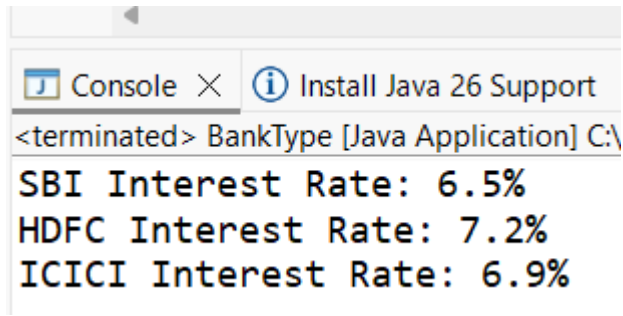


2. Create an abstract class Bank with method `getInterestRate()` and implement it in different bank classes.

Code:

```
abstract class Bank {  
    abstract double getInterestRate();  
}  
  
class SBI extends Bank {  
    public double getInterestRate() {  
        return 6.5;  
    }  
}  
  
class HDFC extends Bank {  
    public double getInterestRate() {  
        return 7.2;  
    }  
}  
  
class ICICI extends Bank {  
    double getInterestRate() {  
        return 6.9;  
    }  
}  
  
public class BankType{  
    public static void main(String[] args) {  
        Bank b1 = new SBI();  
        Bank b2 = new HDFC();  
        Bank b3 = new ICICI();  
        System.out.println("SBI Interest Rate: " + b1.getInterestRate() + "%");  
        System.out.println("HDFC Interest Rate: " + b2.getInterestRate() + "%");  
        System.out.println("ICICI Interest Rate: " + b3.getInterestRate() + "%");  
    }  
}
```

Output:



```
<terminated> BankType [Java Application] C:\nSBI Interest Rate: 6.5%\nHDFC Interest Rate: 7.2%\nICICI Interest Rate: 6.9%
```

3. Write a program where an interface is implemented and exception handling is used inside the method.

Code:

```
interface Calculator {  
    void divide(int a, int b);  
}  
  
class CalculatorImpl implements Calculator {  
    public void divide(int a, int b) {  
        try {  
            int result = a / b;  
            System.out.println("Result: " + result);  
        } catch (ArithmeticException e) {  
            System.out.println("Error: Cannot divide by zero");  
        } finally {  
            System.out.println("Operation completed.");  
        }  
    }  
}  
  
public class Main {  
    public static void main(String[] args) {  
        Calculator calc = new CalculatorImpl();  
        calc.divide(10, 2);  
        calc.divide(10, 0);  
    }  
}
```

}

Output:

